



Chapter 4.1

NUMERICAL DIFFERENTIATION

The derivative of the function f at x_0 is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

This formula gives an obvious way to generate an approximation to $f'(x_0)$; simply compute

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

for small values of h . Although this may be obvious, it is not very successful, due to our old nemesis round-off error. But it is certainly a place to start.

Three-Point Endpoint Formula

- $$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0),$$

where ξ_0 lies between x_0 and $x_0 + 2h$.

Three-Point Midpoint Formula

$$g'(x_1) = \frac{1}{2} [g(x_2) - g(x_0)]$$

- $f'(x_0) = \frac{1}{2h} [f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6} f^{(3)}(\xi_1),$

where ξ_1 lies between $x_0 - h$ and $x_0 + h$.

$$\begin{array}{ccc} x_0 & x_1 & x_2 \\ & \uparrow & \end{array}$$

5. Use the most accurate three-point formula to determine each derivative in the following tables.

a.	x	$f(x)$	$f'(x)$
	1.1	9.025013	17.769705
	1.2	11.02318	22.193635
	1.3	13.46374	27.10735
	1.4	16.44465	32.51085

b.	x	$f(x)$	$f'(x)$
	8.1	16.94410	
	8.3	17.56492	
	8.5	18.19056	
	8.7	18.82091	

⑤ a)

3 endpoint (forward) $\circ$ (h is pos)

$$f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0+h) - f(x_0+2h)]$$

$$f'(1.1) = \frac{1}{2(0.1)} [-3f(1.1) + 4f(1.2) - f(1.3)]$$

$$= \frac{1}{2(0.1)} [-3(9.025013) + 4(11.02318) - 13.46374] = 17.769705$$

$$f'(1.2) = \frac{1}{2(0.1)} [-3(11.02318) + 4(13.46374) - 16.44465] = 22.193635$$

3 endpoint (backward) $\circ$ (h is neg)

$$f'(1.3) = \frac{1}{2(-0.1)} [-3(13.46374) + 4(11.02318) - 9.025013] = 26.617565$$

$$f'(1.4) = \frac{1}{2(-0.1)} [-3(16.44465) + 4(13.46374) - 11.02318] = 32.51085$$

3 midpoint formula (h is pos)

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)]$$

$$f'(1.2) = \frac{1}{2(0.1)} [f(1.3) - f(1.1)]$$

$$= \frac{1}{2(0.1)} [13.46374 - 9.025013] = 22.193635$$

$$f'(1.3) = \frac{1}{2(0.1)} [16.44465 - 11.02318] = 27.10735$$

Five-Point Midpoint Formula

$$f'(x_3) = \frac{1}{12h} [f(x_1) - 8f(x_2) + 8f(x_3) - f(x_4)]$$

- $f'(x_0) = \frac{1}{12h} [f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)] + \frac{h^4}{30} f^{(5)}(\xi),$
(4.6)

where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.

The derivation of this formula is considered in Section 4.2. The other five-point formula is used for approximations at the endpoints.

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$
 ↑

Five-Point Endpoint Formula

forward+backward

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$

$$f'(x_0) = \frac{1}{12h} [-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) + 16f(x_0 + 3h) - 3f(x_0 + 4h)] + \frac{h^4}{5} f^{(5)}(\xi),$$

where ξ lies between x_0 and $x_0 + 4h$.

$$x_1 = \frac{1}{12h} [-25f(x_1) + 48f(x_2) - 36f(x_3) + 16f(x_4) - 3f(x_5)]$$

Second Derivative Midpoint Formula

- $$f''(x_0) = \frac{1}{h^2}[f(x_0 - h) - 2f(x_0) + f(x_0 + h)] - \frac{h^2}{12}f^{(4)}(\xi), \quad (4.9)$$

for some ξ , where $x_0 - h < \xi < x_0 + h$.

If $f^{(4)}$ is continuous on $[x_0 - h, x_0 + h]$ it is also bounded, and the approximation is $O(h^2)$.

9. Use the formulas given in this section to determine, as accurately as possible, approximations for each missing entry in the following tables.

a.	x	$f(x)$	$f'(x)$
$h=0.1$	$2.1 \ x_0$	-1.709847	3.8993
	$2.2 \ x_1$	-1.373823	2.8769
	$2.3 \ x_2$	-1.119214	2.2497
	$2.4 \ x_3$	-0.9160143	1.8378
	$2.5 \ x_4$	-0.7470223	1.5442
	$2.6 \ x_5$	-0.6015966	1.3555

5 point forward end point methods (needs 4 po. - a)

$$f'(x_0) = \frac{1}{12h} [-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) + 16f(x_0 + 3h) - 3f(x_0 + 4h)]$$

$$f'(2.1) = \frac{1}{12(0.1)} [-25(-1.709847) + 48(-1.373823) - 36(-1.119214) + 16(-0.9160143) - 3(-0.7470223)] = 3.8993$$

$$f'(2.2) = \frac{1}{12(0.1)} [-25(-1.373823) + 48(-1.119214) - 36(-0.9160143) + 16(-0.7470223) - 3(-0.6015966)] = 2.8769$$

5 point midpoint (needs 2 points forward and 2 points backward)

$$f'(x_0) = \frac{1}{12h} [f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)]$$

$$f'(2.3) = \frac{1}{12(0.1)} [-1.709847 - 8(-1.373823) + 8(-0.9160143) - (-0.7470223)] = 2.2497$$

$$f'(2.4) = \frac{1}{12(0.1)} [-1.373823 - 8(-1.119214) + 8(-0.7470223) - (-0.6015966)] = 1.8378$$

5 point backward endpoint method (needs 4 points backwards and h is negative)

$$f'(x_0) = \frac{1}{12h} [-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) + 16f(x_0 + 3h) - 3f(x_0 + 4h)]$$

$$f'(2.5) = \frac{1}{12(0.1)} [-25(-0.7470223) + 48(-0.9160143) - 36(-1.119214) + 16(-1.37823) - 3(-1.709847)] = 1.5442$$

$$f'(2.6) = \frac{1}{12(0.1)} [-25(-0.6015966) + 48(-0.7470223) - 36(-0.9160143) + 16(-1.119214) - 3(-1.37823)] = 1.3555$$

Exercise Set 4.1

1. Use the forward-difference formulas and backward-difference formulas to determine each missing entry in the following tables.

a.

x	$f(x)$	$f'(x)$
0.5	0.4794	0.852
0.6	0.5646	0.852
0.7	0.6442	0.796

b.

x	$f(x)$	$f'(x)$
0.0	0.00000	
0.2	0.74140	
0.4	1.3718	

2. Use the forward-difference formulas and backward-difference formulas to determine each missing entry in the following tables.

a.

x	$f(x)$	$f'(x)$
-0.3	1.9507	
-0.2	2.0421	
-0.1	2.0601	

b.

x	$f(x)$	$f'(x)$
1.0	1.0000	
1.2	1.2625	
1.4	1.6595	

3. The data in Exercise 1 were taken from the following functions. Compute the actual errors in Exercise 1, and find error bounds using the error formulas. *only*

a. $f(x) = \sin x$

b. $f(x) = e^x - 2x^2 + 3x - 1$

3. The data in Exercise 1 were taken from the following functions. Compute the actual errors in Exercise 1, and find ~~error bounds~~ ^{only} using the error formulas.

a. $f(x) = \sin x$

b. $f(x) = e^x - 2x^2 + 3x - 1$

$$f'(x) = \cos x$$

* Note: put
radian on
calculator

$$f'(0.5) = \cos(0.5) = 0.87758$$

$$\text{Actual error: } |0.877582 - 0.852| = 0.025$$

$$f'(0.6) = \cos(0.6) = 0.825$$

$$\text{Actual error: } |0.825 - 0.852| = 0.027$$

$$f'(0.7) = \cos(0.7) = 0.7648$$

$$\text{Actual error: } |0.7648 - 0.796| = 0.0312$$

Exercise Set 4.1

4. The data in Exercise 2 were taken from the following functions. Compute the actual errors in Exercise 2, and find error bounds using the error formulas.

a. $f(x) = 2 \cos 2x - x$

b. $f(x) = x^2 \ln x + 1$

5. Use the most accurate three-point formula to determine each missing entry in the following tables.

a.

x	$f(x)$	$f'(x)$
1.1	9.025013	
1.2	11.02318	
1.3	13.46374	
1.4	16.44465	

b.

x	$f(x)$	$f'(x)$
8.1	16.94410	
8.3	17.56492	
8.5	18.19056	
8.7	18.82091	

c.

x	$f(x)$	$f'(x)$
2.9	-4.827866	
3.0	-4.240058	
3.1	-3.496909	
3.2	-2.596792	

d.

x	$f(x)$	$f'(x)$
2.0	3.6887983	
2.1	3.6905701	
2.2	3.6688192	
2.3	3.6245909	

Exercise Set 4.1

7. The data in Exercise 5 were taken from the following functions. Compute the actual errors in Exercise 5, and find error bounds using the error formulas.

a. $f(x) = e^{2x}$

b. $f(x) = x \ln x$

c. $f(x) = x \cos x - x^2 \sin x$

d. $f(x) = 2(\ln x)^2 + 3 \sin x$

a) $f'(x) = 2e^{2x}$

$$f'(1.1) = 2e^{2(1.1)} = 18.050027$$

$$17.769705$$

Actual error:

$$|18.050027 - 17.769705| = 0.280322$$

$$f'(1.2) = 2e^{2(1.2)} = 22.046353$$

$$22.193635$$

$$|22.046353 - 22.193635| = 0.147282$$

$$f'(1.3) = 2e^{2(1.3)} = 26.927476$$

$$27.10735$$

$$|26.927476 - 27.10735| = 0.179874$$

$$f'(1.4) = 2e^{2(1.4)} = 32.82929$$

$$32.51085$$

$$|32.82929 - 32.51085| = 0.31844$$

Ex 4.1
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5-point method

x	f(x)
2.1	-1.709847
2.2	-1.373823
2.3	-1.119214
2.4	-0.9160143
2.5	-0.7470223
2.6	-0.6015966

f'(x)

2.5 = x₀ + h
2.5 = x₀

midpoint

$$f'(2.3) = \frac{1}{12(0.1)} [-1.709847 - 8(-1.373823) + 8(-0.9160143) - (-0.7470223)]$$

$$f'(2.4) = \frac{1}{12(0.1)} [-1.373823 - 8(-1.119214) + 8(-0.7470223) - (-0.6015966)]$$

Backward

$$f'(2.5) = \frac{1}{12(0.1)}$$

5-point endpoint

$$[-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)]$$

Backward (h < 0)

$$f'(2.5) = \frac{1}{12(0.1)} [-25(-0.7470223) + 48(-0.9160143) - 36(-1.119214) + 16(-1.373823) - 3(-1.709847)]$$

$$f'(2.6) = \frac{1}{12(0.1)} [-25(-0.6015966) + 48(-0.7470223) - 36(-0.9160143) + 16(-1.119214) - 3(-1.373823)]$$